Physics 1 Test

Chapters 3 and 4: Forces and motion

Name	______________________
Date	Thursday 15 October 2026
Time	90 minutes
Aids	Part 1: none. Part 2: all, except communication

Answer all the exercises. Show your working and justify your answers; a correct answer without justification does not earn full credit. Use figures where helpful.

Part 1 is handed in before you begin Part 2. The test has 5 exercises and is worth 20 points in total.

Part 1

Without aids

Exercise 1 – Forces on an incline (4 points)

A block of mass m rests on an incline making an angle θ with the horizontal.

a) Draw a figure and mark all the forces acting on the block.

b) Show that the normal force is $N = mg\cos\theta$.

c) How large must the friction force be, at minimum, for the block to stay at rest?

Exercise 2 – Equations of motion (4 points)

A ball is thrown straight up with speed $v_0 = 12\,\mathrm{m/s}$. Neglect air resistance, and use $g = 9.8\,\mathrm{m/s^2}$.

a) How high does the ball go?

b) How long does it take before the ball is back at its starting point?

Exercise 3 – Vectors (2 points)

Two forces $\vec{F}_1$ and $\vec{F}_2$ act at the same point. $\vec{F}_1$ has magnitude $3.0\,\mathrm{N}$ and points east; $\vec{F}_2$ has magnitude $4.0\,\mathrm{N}$ and points north. How large is the resultant force?

A $1.0\,\mathrm{N}$

B $5.0\,\mathrm{N}$

C $7.0\,\mathrm{N}$

D $12\,\mathrm{N}$

Part 2

With aids

Exercise 4 – The braking car (6 points)

A car is travelling at 80 km/h when the driver notices an obstacle 50 m ahead. The reaction time is 0.8 s, and the brakes give a constant deceleration of $6.5\,\mathrm{m/s^2}$.

a) How far does the car travel during the reaction time?

b) Does the car manage to stop before the obstacle? Show your working.

The figure below shows a velocity–time graph of the motion.

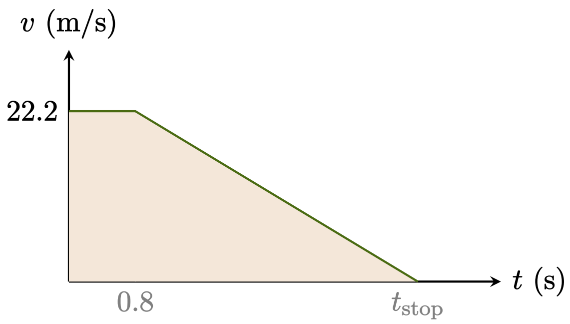

Figure 1. Velocity as a function of time. The area under the graph is the distance.

c) Use figure 1 to explain how the stopping distance can be read off the graph.

Exercise 5 – Numerical solution (4 points)

The program below simulates the fall of a ball with air resistance using Euler's method.

```python
import numpy as np

m, g, k = 0.10, 9.81, 0.02      # mass, g, drag
v, t, dt = 0.0, 0.0, 0.01

while t < 5.0:
    a = g - (k / m) * v**2      # acceleration (m/s^2)
    v = v + a * dt
    t = t + dt

print(f"Speed after {t:.1f} s: {v:.2f} m/s")
```

a) Explain what line 7 does, and which physical law it is based on.

b) What will the program print if k is set to zero? Justify your answer without running the program.

c) Change the program so that it also computes how far the ball has fallen.

Good luck!